

NH3Bench: Language Agents as Real-Time Operators of an Industrial Plant

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<https://github.com/nicl-nno/NH3Bench>

Abstract

We demonstrate NH3Bench, an executable benchmark for a general question: can a language-model agent hold real-time supervisory control of an industrial installation? Three properties define the task and are absent from digital-native agent benchmarks: the process evolves whether or not the agent acts (inaction ends in catastrophe); it does not pause for deliberation (reasoning tokens convert to virtual seconds, so a correct answer can arrive after the rupture); and it is only partially digitalized (part of the truth lives on local gauges, fetched by dispatched workers). NH3Bench instantiates it on a physics-based digital twin of an industrial ammonia (R717) refrigeration plant; the agent is the shift engineer. Across six calibrated scenarios, four Claude-family models score between 32 and 91 of 100. Attendees watch live episodes with the models' reasoning streams, then play the same scenarios under the same clock in a browser trainer running the same twin.

Introduction

Language-model agents are being proposed for operational roles in physical industries. Beneath them lies a general question: can an agent hold *real-time supervisory control* of a running industrial installation—keep a continuous physical process inside its safe envelope using the plant's instruments, a fixed set of actuators and safety systems, and its human workforce, on the plant's own clock?

Existing agent benchmarks evaluate digital-native surrogates of this task: tool use in software environments (Liu et al. 2023), customer-service dialogues under rule compliance (Yao et al. 2024), procedure following with the correct procedure supplied (Nandi et al. 2025). Industrial-asset suites stop short of a closed control loop (Patel et al. 2025), and real-time operator studies use abstract text simulators without physics (Lee et al. 2026). The general task differs from all of these in three structural properties. The dynamics are autonomous: in a developing incident *doing nothing is itself a decision* with physical consequences, so there is no safe default from which to grade only errors of commission. Deliberation is not free: the plant integrates forward while the agent thinks, so correctness and latency cannot be scored separately. And no SCADA sees everything: the information set includes sight glasses, local gauges, smells and sounds,

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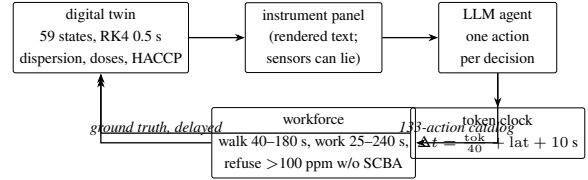


Figure 1: The NH3Bench loop. The agent sees rendered instrument readings, not plant state, and answers with one action id from a fixed 133-action catalog: remote commands, safety systems, or work orders that dispatch one of two simulated operators. Manual measurements return ground truth, bypassing failed sensors—in most scenarios SCADA disagrees with reality at least once. Every reasoning token advances the plant by 1/40 s before the action executes, so verbosity is a safety variable. A demonstration video will be available at the repository URL above.

extended only by dispatching people at a cost in minutes and, sometimes, exposure. A benchmark for the task must operationalize all three; none did.

NH3Bench is an executable instantiation of the task. The three properties only bite when consequences come from physics, not script, so instantiation requires a concrete plant. Our case study is industrial ammonia (R717) refrigeration, chosen for its documented accident record and its rich split between digital and analog truth: the 2010 Millard Refrigerated Services ammonia release was caused by a defrost cycle interrupted by a power loss (U.S. Chemical Safety and Hazard Investigation Board 2015)—the mechanism reconstructed in our first scenario. Every outcome is produced by a thermodynamic model whose mass balance closes to under 0.1 % and which reproduces a Millard-class hydraulic shock (201 bar against a 186 bar pipe limit).

Benchmark Design

The task interface is domain-independent: the agent sees rendered instrument readings, answers with one action id from a fixed catalog, and shares its clock with the plant (Figure 1). The ammonia case study fills this interface with physics: the twin models a 250 t/day dairy plant carrying a 4,170 kg R717 charge—four screw compressors on two suction stages, two evaporative condensers, three vessels, six

policy	S1	S2	S3	S4	S5	S6	mean
oracle (scripted)	100	100	100	100	100	100	100.0
Fable 5	100	100	100	100	78	69	91.2
Opus 5	100	100	80	100	67	0	74.6
always-ESD	73	95	57	95	57	0	62.8
Sonnet 5	0	0	80	53	84	69	47.6
written regulation	0	95	80	0	94	0	44.9
Haiku 4.5	0	0	100	0	94	0	32.4
inaction	0	0	80	0	100	0	30.1

Table 1: Unified score, seed 1. Complete per-decision transcripts of all 24 measured model episodes are published in the repository.

air coolers, hot-gas defrost, an ice-water circuit, HACCP-critical product temperatures, and an ammonia dispersion model with per-operator toxic-dose accounting.

Six scenarios are admitted by a calibration quartet plus an always-ESD baseline: inaction must end in catastrophe; a scripted oracle must pass cleanly on observable evidence; a faithful checklist policy (π_{reg}) must fail or pay heavily; random action must not save. Mechanisms include a hung defrost silently swallowing its remote abort (S1), a loud leak masking a dead level sensor (S2), an air-in-condenser false trail (S3), and an isolation trap where the regulation’s own reflex causes the rupture (S4). Two scenarios are *mirrored controls*: S5 punishes over-reaction to a drifting gas detector, while S6 presents the same evidence picture and then lets the discredited instrument start telling the truth—the fatal error is closing the question and never re-opening it. S6 also closes an empirical loophole: “emergency-stop everywhere” scores 88.6/100 on S1–S5 with no diagnosis at all, but ends in catastrophe on S6, whose leak is independent of the compressors.

The metrics, too, attach to the task, not the domain: alongside Prevention Rate the benchmark reports *Regulation Gap*—score minus that of the published checklist policy; an agent that cannot beat the written regulation scores below zero regardless of raw prevention. A unified 0–100 score multiplies independent axes: catastrophe or an incapacitated worker zeroes the run; operator dose multiplies (never monetized); economics and discipline fill the remainder. Points of no return, found by bisection over delayed-oracle starts, separate “did not understand” from “understood too slowly.”

Measured Results

Table 1 summarizes four Claude-family models against the reference policies. Three findings organize the demonstration. (1) *The token clock dominates the small model*: Haiku 4.5 spends $\sim 4,300$ tokens per decision, gets five decisions in S1’s 614-second window, and loses scenarios it demonstrably understood; in S6 it reached the correct decision—evacuate—and spent 6,598 tokens (165 virtual seconds) writing it down while the gland let go. (2) *Frontier failures are dispositional, not epistemic*: every model found the correct answer in S5 (the stronger the model, the earlier), and every model kept escalating afterwards; rankings on the mirrored pair S5/S6 invert, so no fixed disposition survives both.

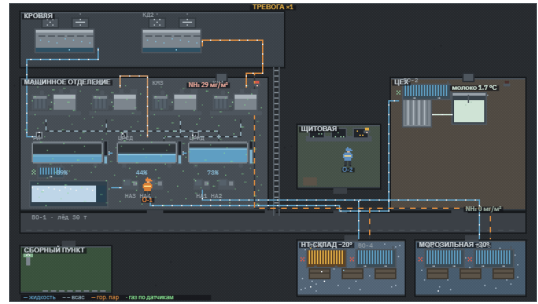


Figure 2: The trainer’s plant view 33 minutes into S2 of the ammonia case study: the machine-room NH_3 detector reads 29 mg/m^3 , the alarm banner flashes, the drum level sensor reads a quietly impossible 73 %, and operator O-1 walks to the local level glass—the only way left to learn the truth. The map renders only agent-visible state (labels in Russian, the validating operator’s language).

(3) *Choreography is a capability*: Opus 5 diagnosed S6 correctly and ordered the correct isolation three times—one cancelled by its own evacuation order, one refused because the breathing apparatus was on the wrong operator, one 232 seconds too late. The four-model, 24-episode matrix is valued at $\approx \$244$ at list prices ($\$0.3\text{--}37$ per episode).

Demonstration

After a one-minute walkthrough of the plant and action catalog, the demonstration has three phases:

1. **Watch**: the attendee picks a model and a scenario and follows the live episode—panel, clock, work orders, and the model’s own reasoning stream, with signature moments like dying mid-thought holding the right answer.
2. **Play**: a browser trainer (Figure 2), built for expert validation by a practicing ammonia-plant operator, executes the *same* digital twin via Pyodide—no installation, physics identical by construction—and the attendee runs the same scenario under the same clock. Their unified score joins a live leaderboard seeded with Table 1; the mirrored pair S5/S6 is as unforgiving of human dispositions as of model ones.
3. **Inspect**: annotated per-decision transcripts of the measured episodes, aligned with the plant trajectory.

Conclusion

NH3Bench turns a general question—can a language agent hold real-time supervisory control of an industrial installation?—into a measurable one. Its capability triad belongs to the task, not to ammonia: acting under irreversible physical consequences, budgeting one’s own deliberation, buying information with human exposure. The recipe—calibrated twin, token clock, workforce layer, accident-forcing admission, mirrored controls—transfers to any installation with a faithful twin; the ammonia case study is one such point, cheap enough to re-run at every model release. All code, scenarios, reference policies, metrics and transcripts are public at the repository above.

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